

CS6360: Assignment 1

A Theoretical Result / Mathematical Problem Formulation

Static and Dynamic System

S No.	Parameter	Static System	Dynamic System
1.	Explanation	System does not change over time.	System changes over time.
2.	I/O Relationship	Fixed Input Output Relationship	Time Varying Input Output Relationship
3.	Representation	Represented by algebraic equations	Represented by differential equations.
4.	Time Invariance	Behaviour does not change over time.	Behaviour changes over time.
5.	Behaviour	Linear and predictable pattern	Non Linear and non predictable pattern



Julen Urain 

4 years ago

Very elegant approach. It would be nice to point out why the Hamiltonian approach is better for modelling in comparison with the baseline



Background

Hamiltonian neural networks

[S Greydanus](#), [M Dzamba](#)... - Advances in neural ..., 2019 - proceedings.neurips.cc

... In this paper, we draw inspiration from **Hamiltonian** mechanics, a branch of physics ... of crafting the **Hamiltonian** by hand, we propose parameterizing it with a **neural network** and ...

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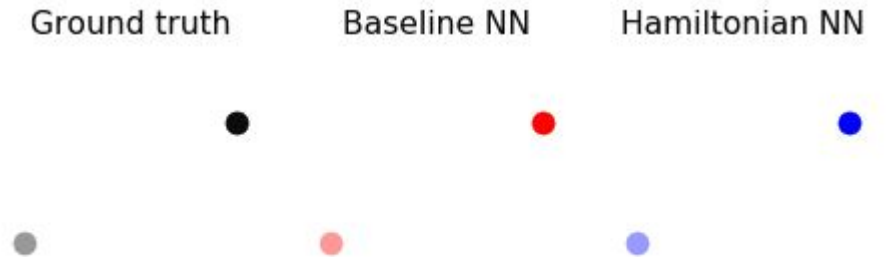
Hamiltonian Neural Networks

Sam Greydanus, Misko Dzamba, and Jason Yosinski (**NeurIPS 2019**)

Abstract

- In this paper, authors take inspiration from Hamiltonian mechanics to train models that learn and respect exact conservation laws in an unsupervised manner. They evaluate the model on problems where conservation of energy is important. Our model trains faster and generalizes better than a regular neural network.

Core idea: *Instead of crafting Hamiltonians by hand, we parameterize them with neural networks and then learn them directly from data.*



Two bodies interact via a gravitational force. The dynamics of the baseline model do not conserve total energy and quickly diverge from ground truth. Meanwhile, the HNN learns to conserve a quantity that is close to total energy. This makes its predicted trajectories more stable.

Why HNN?

- The first problem is its **notion of discrete “time steps”** that connect neighboring states. Since time is actually continuous, a better approach would be to express dynamics as a set of differential equations and then integrate them from an initial state at t_0 to a final state at t_1 .
- The second problem with existing methods is that they tend not to learn exact conservation laws or invariant quantities. This often causes them to **drift away from the true dynamics** of the system as small errors accumulate.
 - **Techniques like Neural ODEs take a step in the right direction**

General scenario.

- Fits well for physical systems. But what about non physical dynamical systems?

Framework

Port-Hamiltonian Neural Networks for Learning Explicit Time-Dependent Dynamical Systems

- Shaan A. Desai, Marios Mattheakis, David Sondak, Pavlos Protopapas, and Stephen J. Roberts, Phys. Rev., 2021

Hamiltonian or Lagrangian neural networks predicts trajectories of physical systems. These networks generally tackle **autonomous systems** that depend implicitly on time or systems for which a control signal is known apriori.

Real world dynamical systems are **non-autonomous**, driven by time-dependent forces and experience energy dissipation. Hence, this challenge of learning from such non-autonomous systems is addressed by embedding the **port-Hamiltonian framework** that can capture energy dissipation and time-dependent control forces.

They show that the proposed **port-Hamiltonian neural network** can efficiently learn the dynamics of nonlinear physical systems of practical interest and accurately recover the underlying stationary Hamiltonian, time-dependent force, and dissipative coefficient.

Compositional Learning

Compositional Learning of Dynamical System Models Using Port-Hamiltonian Neural Networks

- Cyrus Neary and Ufuk Topcu, Proceedings of Machine Learning Research 2023

Dynamical systems: Composed of interconnected subsystems.

- 2) A framework for compositional neural networks (**CoNN**).
- 3) Algorithms to train these models.
- 4) A method to compose the **learned** models (possibly pretrained models).
- 5) Theoretical results that bound the error of the resulting composite models,
- 6) A method to learn the composition itself when it is not known a priori.

End result is a modular approach to compositional learning.

- Neural network submodules are trained on trajectory data generated by relatively simple subsystems.
- Dynamics of complex composite systems are then predicted without requiring additional data generated by system.

Problem Statement

"The meaning of a compound expression is a function of the meanings of its parts and of the way they are syntactically combined."

- Barbara Partee

'Interconnectivity' or **'compositionality'** of a complex, large-scale, systems can be built up from simpler parts, and that certain **properties** of the complex system can be studied in terms of its constituent **parts** and the way they are **interconnected**.

$$Y = w_0 + w_1X + w_2X^2$$

$$Y = L + NL$$

$$Y = F(L, +, NL)$$

$$\text{Estimate: } Y = F(y_1, y_2)$$

Problem of Compositionality

Q: How to build or learn composite models of dynamical systems from data ?

A: Compositionality is achieved by representing the system of interest, as well as each of its subsystems, as a **port-Hamiltonian neural network (PHNN)**—a class of **neural ordinary differential equations** that uses the port-Hamiltonian systems formalism.

PHNNs parametrize each subsystem's Hamiltonian function, as well as how it dissipates energy, interacts with other subsystems, and how it responds to control inputs.

Composite PHNNs enjoy properties of port-Hamiltonian systems.

Accurate compositions are learned using an amount of training data that is negligible in comparison with that required to train a new model from scratch.

Theory

Port-Hamiltonian Systems Theory

The theory underlying the compositionality of port-Hamiltonian systems concerns the composition of Dirac structures.

port-Hamiltonian systems completely fit within this paradigm, in the sense that the **power conserving** interconnection property of port-Hamiltonian systems defines another port-Hamiltonian system.

$$\text{pH1} + \text{pH2} \rightarrow \text{pH3}$$

It can be extended to infinite-dimensional Dirac structures and port-Hamiltonian Systems

Interactions between the subsystems are captured by the ***interconnection structure*** of the composite system, which may be known a priori, or may itself be learned from data.

Problem

Consider **mass-spring system** as the interconnection of two subsystems, both of which store energy, namely the spring system storing potential energy and the mass system storing kinetic energy.

Power conserving interconnection element by the geometric notion of a Dirac structure, and by adding, next to the energy-storing elements, also energy-dissipating elements.

Dirac structure provides the 'wiring' for the overall system.

Why?

- Modelling electromechanical systems is highly challenging.
- Need expressive, reliable, and well generalizing models.
- [DAE-PINN: A Physics-Informed Neural Network Model for Simulating Differential-Algebraic Equations with Application to Power Networks](#)
- [Efficient and Accurate Gradients for Neural SDEs](#)
- Neural SDEs as Infinite-Dimensional GANs
- NDAE: Neural Differential Algebraic Equations

Interaction term

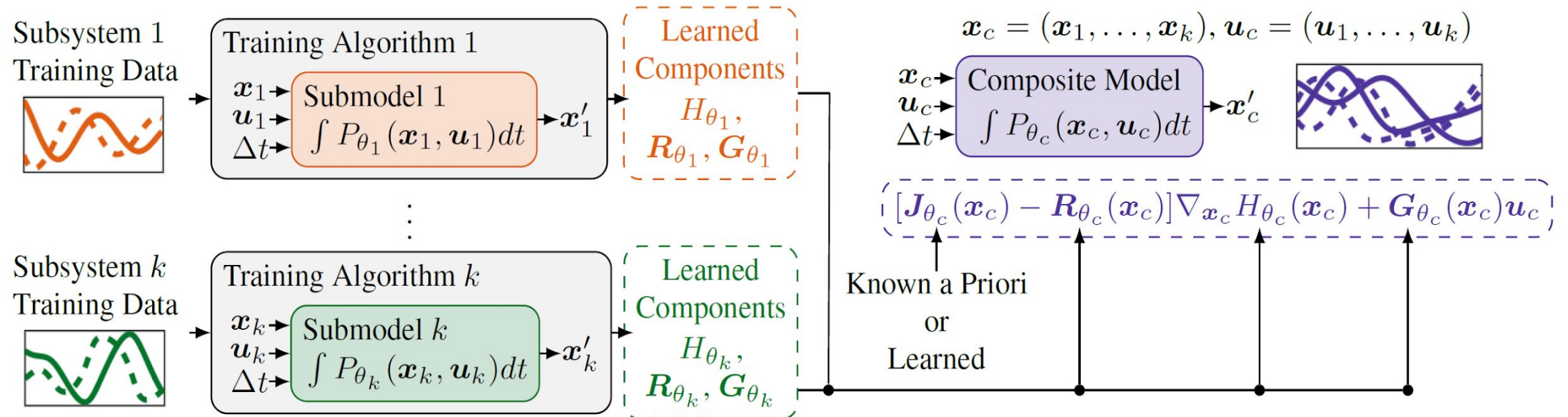
- The framework models individual subsystems independently, and uses physics informed interfaces between these sub models to capture their interactions.
- Submodels are trained on trajectory data generated by relatively simple subsystems.
- Interactions between the subsystems are captured by the interconnection structure of the composite system, which may be known a priori, or may itself be learned from data.
- In many cases this interconnection structure is given by a constant linear operator; we accordingly present a method to **learn it via linear regression**. In the general case, it may instead be parameterized using a neural network.
- **Challenge is learning or estimating the unknown interconnection structure**

Compositional learning framework

- Hamiltonian, dissipation term, and control input term

Train separate (PHNNs) on data generated by individual subsystems.

- Compose these sub models to construct another PHNN that models
- the composite system



Roadmap

- HNN
- Port HNN
- incremental PHNN
- Monotonicity